

A7 Executing through Terminal - Answers

Python - Executing Programs Through Terminal

1. Which command is commonly used to execute a Python file named program.py from the terminal?

- o A) run program.py
- o B) python program.py
- o C) execute program.py
- o D) start program.py

Answer: B) Run it with python program.py.

2. On systems where python points to Python 2 or is unavailable, which command is commonly used to run a Python 3 program?

- o A) python3 program.py
- o B) py3 program.py
- o C) run3 program.py
- o D) execute3 program.py

Answer: A) python3 program.py explicitly invokes Python 3.

3. Consider the following Python file named hello.py:

```
print("Hello World")
```

What command should be used to execute it from the terminal?

- A) python hello
- B) python hello.py
- C) execute hello.py
- D) run hello.py

Answer: B) python hello.py.

4. What does the following command do?

```
python test.py
```

- A) Creates a Python file called test.py
- B) Opens test.py for editing
- C) Executes test.py using Python
- D) Deletes test.py

Answer: C) The command executes test.py using Python.

5. A Python file is located at:

```
C:\Projects \ML\model.py
```

Which command can execute it from the terminal?

- A) python C: \Projects \ML\model.py
- B) run C: \Projects \ML\model.py

- C) execute C: \Projects \ML\model.py
- D) start -python C: \Projects \ML\model.py

Answer: A) Use python followed by the full path; quote paths containing spaces. On Windows, python "C:\Projects\ML\model.py".

6. What is the purpose of the cd command in a terminal?

- o A) Create a directory
- o B) Delete a directory
- o C) Change the current directory
- o D) Compile Python code

Answer: C) cd changes the current directory.

7. Suppose the terminal is currently in:

C:\Users \Student

and the Python file is located in:

C:\Users \Student \Projects

Which command can be used to move into the Projects directory?

- A) cd Projects
- B) move Projects
- C) goto Projects.py
- D) open Projects

Answer: A) cd Projects, relative to the current directory shown.

8. Consider the following sequence of commands:

cd Documents

python program.py

What does the second command assume?

- A) program.py exists in the Documents directory
- B) Python is installed inside Documents
- C) Documents is a Python package
- D) program.py must be a module

Answer: A) It assumes program.py exists in Documents (or provide its path).

9. What is the purpose of the command:

python --version

- A) Execute Python code
- B) Display the installed Python version
- C) Update Python
- D) Remove Python

Answer: B) It displays the installed Python version.

10. Which command starts Python's interactive interpreter in a terminal?

- A) python

- B) python.start()
- C) start python.py
- D) interactive python

Answer: A) Running python starts the interactive interpreter when that command is available.

11. If the terminal displays:

>>>

what does this generally indicate?

- A) Python is waiting for interactive input
- B) Python has crashed
- C) The terminal is closed
- D) A Python file is being downloaded

Answer: A) The >>> prompt means the interactive interpreter is waiting for input.

12. What will happen when the following command is executed?

python

- A) A new Python file is created
- B) The Python interactive interpreter starts
- C) All Python programs are executed
- D) Python is installed automatically

Answer: B) It starts the Python interactive interpreter; it does not create a file.

13. Which command exits the Python interactive interpreter?

- A) exit()
- B) stop_python()
- C) close_python()
- D) quit_python()

Answer: A) exit() exits the interactive interpreter (quit() also works).

14. Consider a file:

```
# calculator.py
```

```
a = 10
```

```
b = 20
```

```
print(a + b)
```

What will be displayed after executing:

```
python calculator.py
```

- A) 10
- B) 20
- C) 30
- D) 1020

Answer: C) The script prints 30.

15. What is the likely reason for the error:

python: can't open file 'program.py'

- A) Python cannot perform addition
- B) The file cannot be found at the specified location
- C) Python is always case -sensitive
- D) The terminal does not support Python

Answer: B) Python cannot find the script at the specified path/name.

16. A student runs:

python program.py

and gets:

'python' is not recognized as an internal or external command

What is a likely cause?

- A) The Python interpreter is not installed or is not available in PATH
- B) The .py file is too large
- C) The program contains a loop
- D) Python does not support terminal execution

Answer: A) Python may not be installed or its executable may not be on PATH.

17. Which command can be used to install a Python package from the terminal?

- A) python install numpy
- B) pip install numpy
- C) install python numpy
- D) package numpy

Answer: B) pip install numpy (prefer python -m pip install numpy to target a specific interpreter).

18. Which command can execute a Python module from the terminal?

- A) python -m module_name
- B) python --module module_name
- C) run -m module_name
- D) execute module_name

Answer: A) python -m module_name runs a module as a script.

19. Consider a Python program requiring a command -line argument:

import sys

print(sys.argv[1])

Which command correctly executes it with the argument Hello?

- A) python program.py
- B) python program.py Hello
- C) python Hello program.py
- D) python --arg Hello program.py

Answer: B) python program.py Hello passes Hello as sys.argv[1].

20. What will `sys.argv[0]` generally contain when a Python script is executed from the terminal?

- A) The first user -provided argument
- B) The Python version
- C) The script name/path
- D) The current working directory only

Answer: C) `sys.argv[0]` generally contains the script name or path.

21. Terminal Execution Task

Create a Python file named `calculator.py` containing separate functions for:

`add()`

`subtract()`

`multiply()`

`divide()`

Then execute the program through the terminal using:

`python calculator.py`

The program should accept two numbers from the user and display the results of all four operations.

Restriction: Do not use any external Python libraries.

Answer: Save this as `calculator.py` and run it with `python calculator.py`. It prompts for two numbers and reports all four operations; division by zero is handled explicitly.

```
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return None
    return a / b

def main():
    first = float(input("Enter the first number: "))
    second = float(input("Enter the second number: "))
    print("Addition:", add(first, second))
    print("Subtraction:", subtract(first, second))
    print("Multiplication:", multiply(first, second))
    result = divide(first, second)
    if result is None:
        print("Division: undefined (cannot divide by zero)")
    else:
        print("Division:", result)

if __name__ == "__main__":
    main()
```